



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SARAVANA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

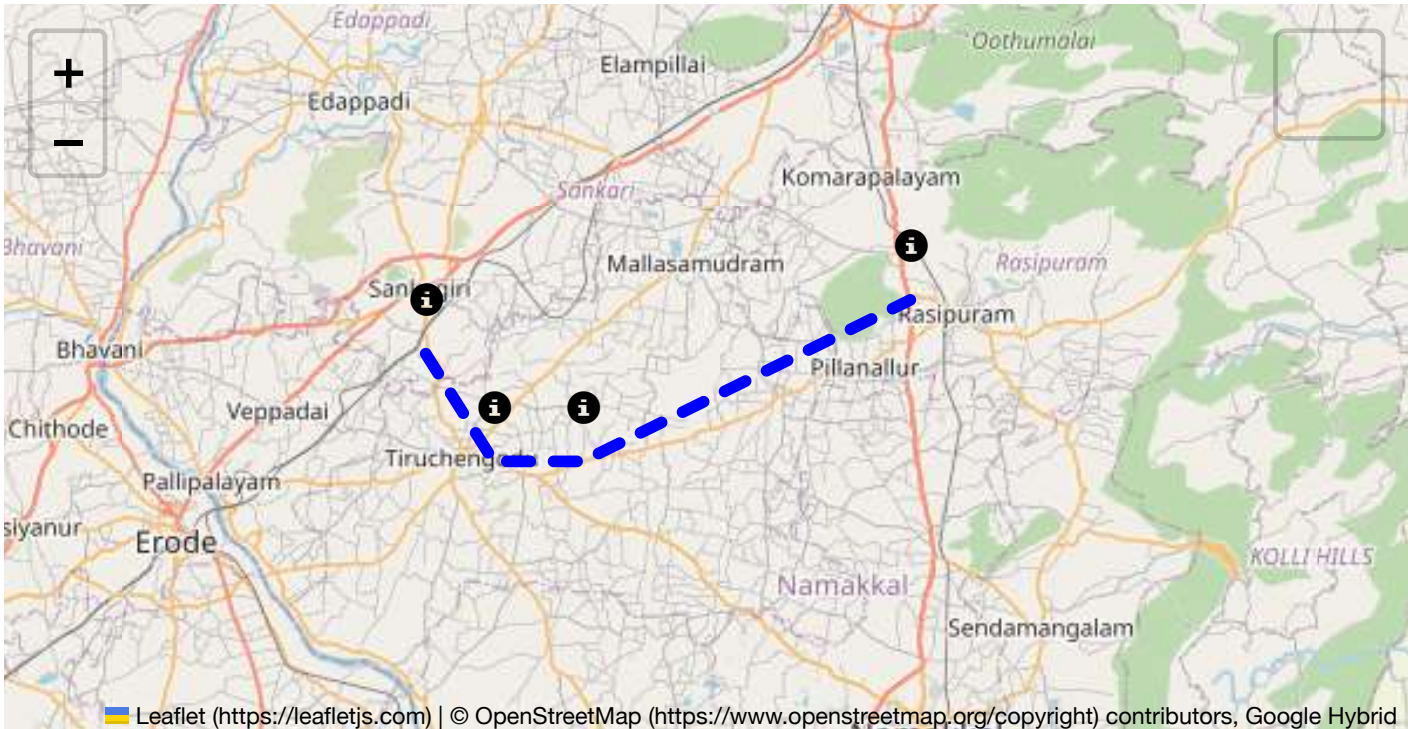
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

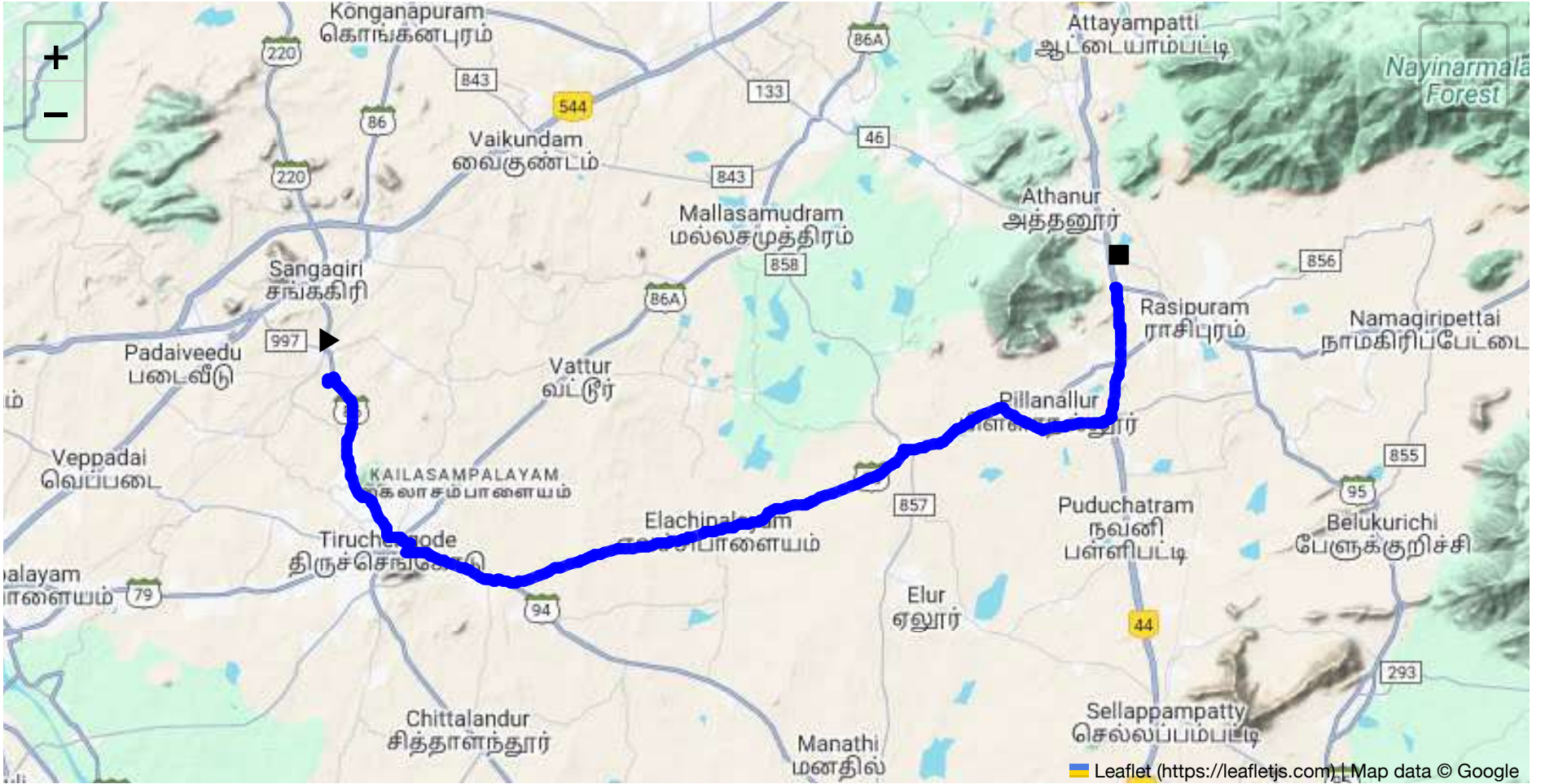
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.45 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours

End: (11.468591, 78.1577)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Rasipuram via Namakkal, primarily following the State Highway 79 (SH 79), spans approximately 44.45 kilometers. It's a moderately busy state highway connecting various rural and semi-urban areas. Expect a journey time of roughly one hour for heavy vehicles transporting hazardous materials.

Weather Conditions

Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The route is likely impacted by:

- **Monsoon rains (June to September):** Risk of reduced visibility and slippery roads.
- **Summer heat (March to May):** Potential overheating issues for vehicles.
- **Occasional cyclones (October to December):** Rare but may cause severe disruptions.

Traffic Patterns

- **Peak Hours:** Mornings (7-10 AM) and evenings (5-8 PM) due to local commuting.
- **Congestion Points:** Namakkal is a significant urban center where congestion could slow traffic, especially near key junctions and market areas.

Road Quality and Infrastructure

- **General Condition:** The highway is relatively well-maintained but can experience wear and tear, particularly after heavy rains.
- **Notable Features:** Some sections may lack adequate lighting and signage.

Alternative Routes

- **During Emergencies:** Consider bypasses through smaller roads connecting directly to the Chennai–Salem Highway for quicker evacuation but carefully verify their capacity and conditions.

Local Regulations

- **Hazardous Material Transport:** Comply with state and national guidelines, including proper placarding. Familiarize yourself with specific local ordinances in each district traversed.

Historical Incidents

- No widely reported incidents specific to this route, but general caution is advised due to incidents elsewhere involving heavy vehicles that required adherence to standard safety protocols.

Environmental Considerations

- **Sensitive Areas:** Nearby agricultural lands; minimize potential spillage and emissions.
- **Wildlife Crossings:** Signage may indicate necessary stops or speed adjustments.

Communication Coverage

- **Potential Dead Zones:** Some rural stretches may have patchy mobile reception. Notably, coverage dips may occur in more isolated areas en route to Kavundampalaiyam.

Emergency Response Times

- **Urban Areas (e.g., Namakkal):** Faster response, approximately 15-20 minutes.
- **Rural Sections:** Response may take upwards of 30-45 minutes depending on accessibility and service availability.

Risk Assessment Summary

- **Overall Risk Level:** Moderate, primarily due to road and weather conditions, with localized congestion points.
- **Recommendations:** Regular vehicular checks, careful monitoring of weather forecasts, and ensuring all communication devices are fully functional to mitigate risks. Engage in defensive driving techniques, especially in congestion-prone areas.

Conclusion

This route, while frequently used and relatively short, requires adherence to safety regulations and cautious navigation to mitigate risks associated with weather, traffic, and general road conditions. Prepare adequately for communication disruptions and ensure contingency plans are in place for emergencies.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	Medium	11.41354, 78.08075	30 KM/Hr
10	Turn	Medium	11.41463, 78.08067	30 KM/Hr
11	Turn	Medium	11.41475, 78.08087	30 KM/Hr
12	Turn	High	11.42964, 78.11587	15 KM/Hr
13	Turn	Medium	11.42945, 78.11606	30 KM/Hr
14	Turn	Medium	11.42940, 78.11609	30 KM/Hr
15	Turn	High	11.42622, 78.15562	15 KM/Hr
16	Turn	Medium	11.43096, 78.15617	30 KM/Hr
17	Turn	Medium	11.46786, 78.15750	30 KM/Hr
18	Turn	High	11.47218, 78.15685	15 KM/Hr
19	Blind Spot	Blind Spot	11.47225, 78.15722	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
10	school	SRV EXCEL Matriculation Higher Secondary School	11.4299679, 78.152616	30 km/h	Medium
11	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
12	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
13	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38190, 77.90254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41354, 78.08075



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41475, 78.08087



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42945, 78.11606



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42940, 78.11609



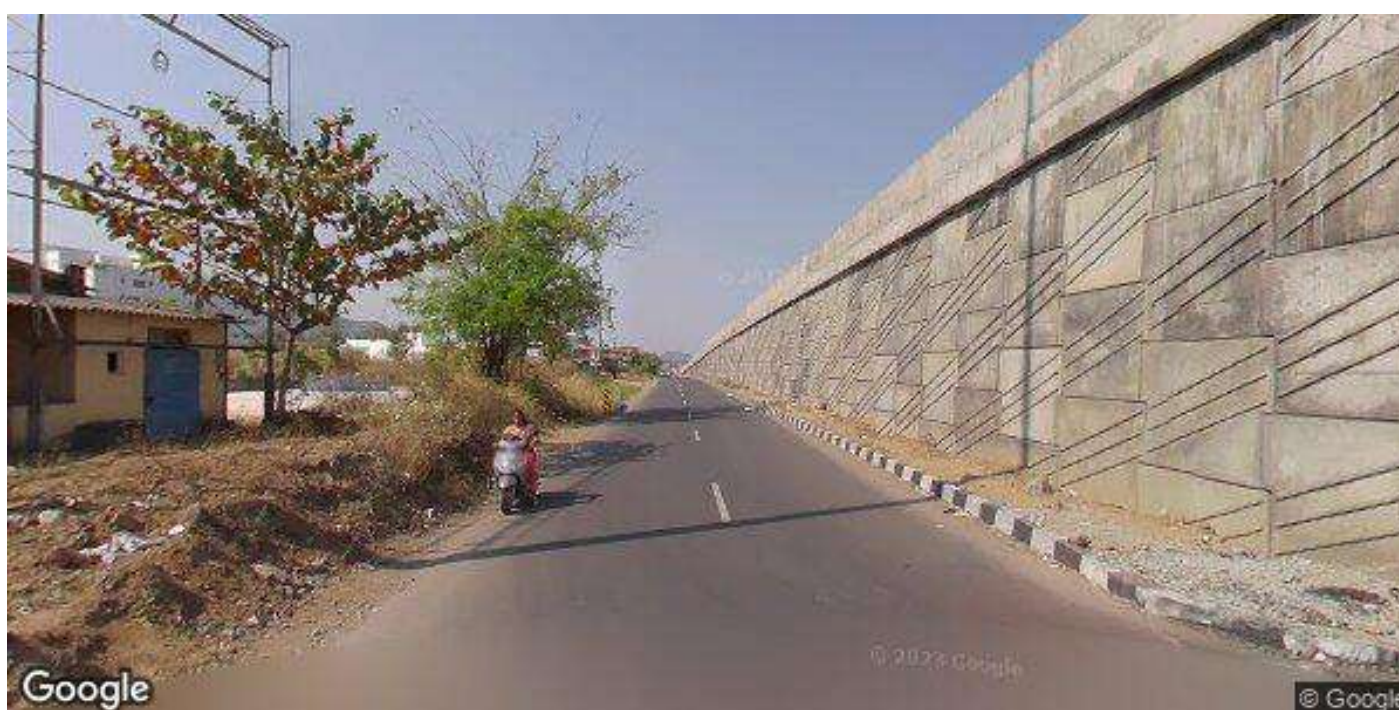
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42622, 78.15562



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43096, 78.15617



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46786, 78.15750



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47218, 78.15685



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.47225, 78.15722
